

A Review of Driving Simulation Technology and Applications

LUCAS BRUCK¹ (Student Member, IEEE), BRUCE HAYCOCK², AND ALI EMADI¹ (Fellow, IEEE)

¹ McMaster Automotive Resource Centre, McMaster University, Hamilton, ON L8P 0A6, Canada

² Toronto Rehabilitation Institute, University of Toronto, Toronto, ON M5G 2A2, Canada

CORRESPONDING AUTHOR: LUCAS BRUCK (e-mail: bruckl@mcmaster.ca).

ABSTRACT Driving simulation has become a very useful tool for vehicle design and research in industry and educational institutes. This paper provides a review of driving simulator components, including the vehicle dynamics model, the motion system, and the virtual environment, and how they interact with the human perceptual system in order to create the illusion of the driving. In addition, a sample of current state-of-the-art vehicle simulators and algorithms are described. Finally, current applications are discussed, such as driver-centered studies, chassis and powertrain design, and autonomous systems development.

INDEX TERMS Driver-in-the-loop, driving simulation, motion cueing, vehicle dynamics, vehicle simulator, virtual simulation.

I. INTRODUCTION

The concept of simulation can be defined as the emulation of a specific behavior through a generic imitating system [1]. In the automotive field, simulation is used both in academia for research and in industry for design purposes. The reason is that by using virtual simulation, new systems can be developed and evaluated within lower time and with lower financial investment. In the automotive industry for instance, virtual simulation can be used throughout all phases of development of a new vehicle. During virtual design, model-in-the-loop analysis enables the comparison of different system architectures, e.g. comparison of powertrain topologies to evaluate the fuel consumption and emissions.

Virtual simulation also speeds the testing phase. Structural, fluid dynamics, and multibody simulation not only optimize the characteristics and geometry of the components, but also work as filters for component selection, e.g. definition of spring pre-load and shock absorber dynamic curve in suspension tuning. Without virtual simulation, not only would the development cost increase, but also the timeframe needed to accomplish each step of development would be longer. During calibration of on-board software parameters or performance components, virtual simulation reduces the number of prototypes that will eventually be manufactured for testing. In this way, the design process as we know it today, with rapid release of new products, is only possible due to simulation. Within

the various types of vehicle simulation, the ability to imitate a ground vehicles response in real time given real driver inputs is called driving simulation and it relies on virtual cues to trick human perception and create the illusion of immersion and motion in a controlled environment.

In the early 1910 s the first motion simulators begin to appear in England and France as a means to safely provide flight training [2]. Motion simulation technology would not be restricted to flight training for long, however. It is not clear though which driving simulator was the first to appear, some researchers acknowledge the system designed in [3], published in 1934 but given the simple architecture of this traffic simulator some authors prefer to point out later works as the real beginning of relevant developments of driving simulators. During the 1970 s several 3 degree-of-freedom (DOF) driving simulators were developed by auto makers and research institutes as depicted in [1], such as the well-known device designed by Volkswagen and another at the Swedish Road and Traffic Research Institute. In addition, the pioneer work to develop the Computer Generated Display Simulator by General Motors together with the Virginia Polytechnic Institute and State University from 1973 to 1975 is acknowledged in [4].

All previous attempts made significant contributions, but it was in 1985 that Daimler-Benz came up with a 6-DOF hexapod motion system integrated to a vehicle dome [5] that would

be later accepted as the most common configuration of high level driving simulators around the globe. Rapid development of new graphic sources happened between the late 1990 s and early 2000s [6]. Digital three-dimensional (3D) graphics, new display resolution, and real-time graphic processing opened a new world of possibilities for driving simulators. The combination of various electronics improvements over the years made the increasing fidelity of driving simulators possible.

Recent studies use driving simulation technology to assess energy management strategies (EMS), human machine interfaces (HMIs), active control systems (ACSS), advanced driver assistance systems (ADAS) and road planning. The purpose of the present work is to break down the mechanisms behind the driving simulation technology, describing the main components of a driving simulator, as a guide for resource centers that want to implement this technology. This work contributes to providing a comprehensive review of the past and current state-of-the-art technologies, driving simulator examples, and applications. Former reviews of this kind can be found in [6], where a thorough survey of relevant simulators at that time was conducted. In that review, the simulators are clustered by their level of cost, a very common form of classification. Additional reviews were presented in [7], with example applications in medicine and engineering, in [8], that details motion system design, and in [9], where the focus is on the classification.

The paper is organized as follows. Section II introduces the human perceptual system. Section III explains the architecture of driving simulators. It discusses each component separately, exposing current technologies and giving examples. Section IV brings examples of driving simulators in research and industry. Section V discusses various applications throughout automotive industry and research. Finally, Section VI concludes the work and state the prospects for driving simulation.

II. HUMAN PERCEPTUAL SYSTEM

Since the role of a driving simulator is to imitate driving in the real world in order to deceive perception, it is important to understand how the human body perceives the driving experience. In driving simulators, multiple systems are combined to form the self-motion perception as further detailed. Because driving is primarily a visual task, one of the most important human sensors to be accounted for in driving simulation is the visual system. In fact, the visual system accounts for the majority of motion perception in a three-dimensional environment [10]. The optical flow together with visual direction and extra-retinal direction form the visual information to be interpreted by the brain in order to define heading [11]. Although less important than the visual system, the auditory system can also be classified as a self-motion perceptual system. It is proven in [12] that through audio means only, separate from other sensory systems, individuals are able to identify, with certain precision, the time-to-collision of sources of noise. Interesting research is conducted in [13] where the authors

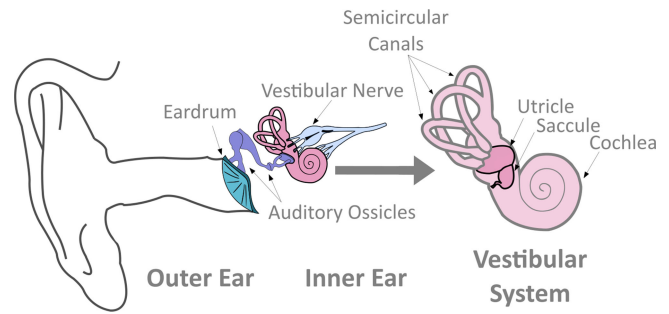


FIGURE 1. Human vestibular system [17].

concluded that adding auditory cues to visual cues increases the illusion of motion.

While in static simulators the visual and auditory systems are responsible for all motion perception, in motion simulators other body sensors are also engaged to build the driving experience. The somatosensory system is responsible for all tactile perceptions and proprioceptive sensors [11], the former accounting for force changes due to motion and the last accounting for position perception and accelerations. As proposed by [14], it is possible to create an illusion of motion by vibrating postural muscles. Therefore, the better the cabin mimics the actual environment of a vehicle cockpit (e.g. by representing with fidelity the internal components, dashboard, seats, texture, commands, and pedals) as well as faithfully representing the vibration and the friction resistance for moving each component, the more realistic the driving experience.

The vestibular system is an inner ear complex comprising semi-circular canals and otolith organs, to recognize linear as well as angular motion [9]. As detailed in [15], the semi-circular canals consist in three circular cavities filled with a fluid (endolymph) that deflect hair lining the canals when an angular acceleration is experienced, such as when the individual is nodding or tilting their head left to right, with an output proportional to angular velocity. The otolith organs, on the other hand are made by the combination of small sacs of sensitive hairs. The role of the sensitive hairs is to sense linear acceleration as well as gravitational forces and transmit it to the central nervous systems. The utricle detects the motion in the horizontal plane, while the saccule detects it in the vertical plane. A representation of the inner ear organs is depicted in Fig. 1. An example of modelling the visual-vestibular system was presented in [16], showing how drivers make use of the vestibular system to determine steering angle. Furthermore, this work provides evidence that drivers adapt depending on how those cues are given.

III. ARCHITECTURE

As mentioned before, the goal of a driving simulator is to create an illusion of the driving experience. To accomplish that, different systems are combined. The diagram depicted in Fig. 2 shows not only the interdependence between each subsystem but also how driver and vehicle model are in the center of the process. Driver input is used to calculate the vehicle

with the driving simulator engine. The work developed in [23] instead builds a platform to co-simulate multiple driving simulators, which increases the realism of the traffic scenario. The study in [24] uses a traffic scenario to evaluate how traffic density affects overtaking and lane change maneuvers.

In [25] the scenario construction was focused on signs and roadside information warnings, instead of other vehicles. The idea was to measure how the driver's journey decision is affected by the information available. Repeatability is very important in early phase of development because engineers can easily visualize the problems a certain system presents in specific condition. Although possible to forecast, weather is usually a limitation in vehicle testing, depending on which region the manufacturer testing center is located. In the case provided by [26] and [27], the low visibility in foggy situations and the risk brought by this condition is evaluated.

C. VISUAL CUES

Over the years, visual cues evolved from analog video presentations and film to digital graphics. These digital graphics rapidly improved from a low number of polygons to high count textured and shaded polygons that provide a highly realistic environment for use in driving simulation. In addition, the improvement of projector technology with increasingly higher pixel resolution, brightness, and contrast ratio, made the accurate projection of such graphics possible [6]. On the contrary, these projectors have been used with similar front projection curved screens as presented in [28] since the 1970 s, which has only now begin to change in limited applications with the use of head mounted devices (HMD). Even with the advent of HMDs, projection screens are still preferable in most cases, as the limited field of view, image lag, and the obscuring of the vehicle interior limit the usefulness of this type of device. Nevertheless, a few authors explored the suitability of these systems in driving simulators in [29] and [30]. Although they foresee an increase in the use of head-mounted devices, they agree that screen-based projection systems still are preferable among tested subjects since they give a better perception of velocity and surroundings.

In an attempt to enhance the visual cues, several authors present different add-on solutions to the conventional rounded screens presented in the most advanced driving simulators. In [31] a visual scale factor is used in an attempt to enhance the speed perception by changing the geometric field of view. In [32], the authors replaced a monoscopic projection system with a 3D-stereoscopic system for improved depth perception, to investigate the use of 3D projector for enhancing velocity as well as distance judgement in a driving simulator. The passive stereoscopic 3D was achieved by using 10 projectors, one projector per eye for each one of the 5 channels for the forward view. The masking is done through a wavelength multiplex process which requires filtering of the projector signals and filtering glasses for the driver. The main drawbacks of such system are the cost associated with the added complexity, the packaging of the projectors in the display structure, and the potential for increased eyestrain and resulting simulator

sickness. By exposing subjects with attested regular stereopsis to a simple car following scenario, the authors were not able to provide strong evidence of the improvements. Lastly in [33], a headlight glare system is created to help understanding how driver performance with different populations is affected by the glare of oncoming vehicles during night driving in rural highway scenarios.

Improvements in PC hardware for rendering virtual worlds has enabled the use of gaming engines in place of the previously used custom rendering hardware. One widely used software package for rendering virtual environments in simulators is OpenSceneGraph (OSG) [34]. Other products that have gained momentum recently given their realistic rendering are Unity [35] and Unreal Engine [36]. Realistic animation and vast object databases can highly improve immersion. The work in [37] assesses how different the driver behavior given different fidelity of visual systems. In that work, a low graphic fidelity simulator was compared to a high fidelity one built with the Unreal engine. The results show that visual attention and situation awareness are highly impacted depending on the fidelity of the visual system. Given its ability to reproduce realistically external objects, Unreal can also be used as a platform to train and test autonomous systems, as developed in [38].

D. AUDITORY CUES

As defined in [39], the importance of the audio cues is justified by how it affects speed judgement, driver awareness, and fatigue. In fact, the audio impact on speed perception is depicted in [40]. In this work, an experiment is conducted with subjects of different age ranges (young and older adults) that are exposed to driving tasks where visual cues to self-motion are provided while the respective presence/absence of auditory cues is manipulated. The conducted experiment proves the assumption that auditory cues affect longitudinal motion perception (speed and acceleration), with older adults more susceptible. Therefore, it is also important to have a well-designed audio library in order to provide quality congruent cues to the driver and maintain the immersion created by the visual system.

Three-dimensional sound is the most suitable choice in driving simulation application [6] since the noise inside the cabin is a combination of noises from different sources, i.e. aerodynamics, tire interaction with the surface, and driveline [9]. As explained in detail in [41], the majority of simulators use the wave table method which consists of lookup tables that receives input values from the sources in the simulation engine (e.g. road, powertrain, and wind sounds) and outputs interpolated cues. In addition, the importance of low frequency speakers is also explained since interior cabin noises are in this frequency range.

E. HAPTIC CUES

The control components (i.e. steering wheel and pedals) act as a bridge between human input and vehicle behavior, making them essential tools in vehicle dynamics feedback. For that

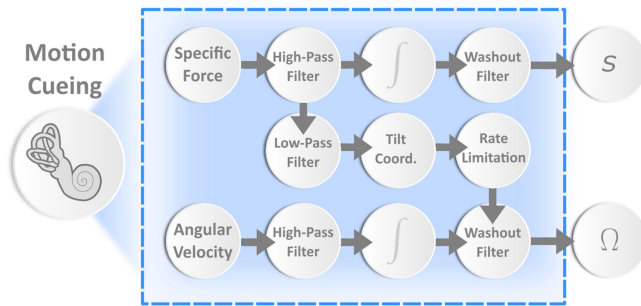


FIGURE 3. Diagram of a classical filter motion cueing algorithm [45].

reason, the human-vehicle interaction must be mimicked in order to provide a high-fidelity driving experience. Most driving simulators include components that provide force feedback through the steering wheel and braking system as exemplified in [1], [6], [42], [43], [39]. The vast majority use a torque motor for the steering wheel force feedback, commanded by the vehicle dynamics model. As stated in [9] the accurate vibration on the steering wheel, mimicking tire-road interaction, provides cues for speed and trajectory, enhancing driver perception.

F. MOTION CUEING ALGORITHM

The presence of a motion system requires the inclusion of a motion cueing algorithm. This algorithm is responsible for creating the displacements of the motion system accounting for human perception and the available workspace [42], therefore its role is to govern the motion of the simulator to provide the driver realistic driving sensations [44]. There are several approaches for motion cueing algorithms as described in the following sections. The work in [45] details the development process of the classical, the optimal, and the adaptive motion cueing for flight simulators, being an essential read for motion cueing developers although it does not include newer approaches such as model predictive control.

Classical Filter: The classical filter (or washout filter) is the most simple and fast method for motion cueing design [46]. As depicted in Fig. 3 the washout algorithm relies on the interaction of high- and low-pass filters that are responsible for the platforms translational and rotational cues [9]. In the picture, S represents the displacement cues and Ω represents the angular. The tilt coordination is a tool used to replicate low frequency sustained accelerations, it performs a tilt of the cabin in order to create a component of the gravitational force in a desired direction [42]. Although the design and implementation of the classical filter is not very complex compared to other methods, its tuning might be time consuming given the fact that it relies extensively on trial and error.

Adaptive Filter: In the adaptive filter algorithm, the parameters that make up the washout algorithm can self-tune. The work in [47] presents a fuzzy logic-based motion cuing classical filter that accounts for both physical boundaries of the platform and error between reference motion and actual

motion when outputting the filter gains. This work develops an algorithm that not only provides a better use of space but also reduces the human perception of motion error, having a performance similar to the algorithms that use model predictive control, later explained. As detailed in [43] the parameters of the motion algorithm can also be calculated through the minimization of a cost function, based on the error between vehicle model and platform acceleration and on the motion system limits. In this case, the sensitivity equations are solved through an extension of Laplace's method (Method of Steepest Descent) which makes the filter to be non-linear. In another work in [48], the authors develop an adaptive tilt-coordination that maximizes the use of the XY environment instead of relying solely on tilt to sustain the acceleration imposed by the driver. The authors accomplish their objective by labeling the acceleration state and regulating the error between the perceived and actual acceleration using linear quadratic regulation. One challenge faced by the authors in this case was the difficulty in finding representative range for the adaptable parameters. The most noticeable issue with using adaptive filters though, is that filter tuning is replaced by cost function tuning. If the parameters are allowed to be updated rapidly, motion distortion resulting in false cues occur.

Optimal Filter: Another approach is the optimal filter algorithm. This method either adds the human perception model (vestibular model) or uses a reference vehicle motion previously recorded to set a comparison between the real and the virtual experience calculated by this new model [49]. The comparison is built and optimized through a transfer function that links the simulator motion inputs to the actual vehicle motion. The objective is to minimize the error in human perception. The major difference between the classical and the optimal approach is the way the filters are calculated, in this case beforehand, through an optimization process that can be done as in [9] through the use of genetic algorithm (GA), or as in [50] using linear quadratic regulator method (LQR) together with GA, or simply the linear quadratic optimization as proposed in [51]. Neural network (NN) approach can also be used as in [52]. The performance of optimal filter methods is linked to the quality of the reference signal whether it comes from a model or empirical measurement. Therefore, model and test limitations can negatively impact the results. Also, the optimization might output different parameters for different set of maneuvers. The impacts of rapidly switching between parameters should be observed as previously mentioned for adaptive filters.

Model Predictive Control Algorithm: Derived from the optimization methods, the model predictive control strategy (MPC) consists of predicting the evolution of dependent variables caused by changes in the independent variables. In essence, the MPC selects an optimal control input at the current time based on prediction of the future behaviour over a given time window. With this future prediction feature the MPC strategy can optimize the motion input, making a better use of the available workspace and motion structure when compared to simple methods such as the classical washout

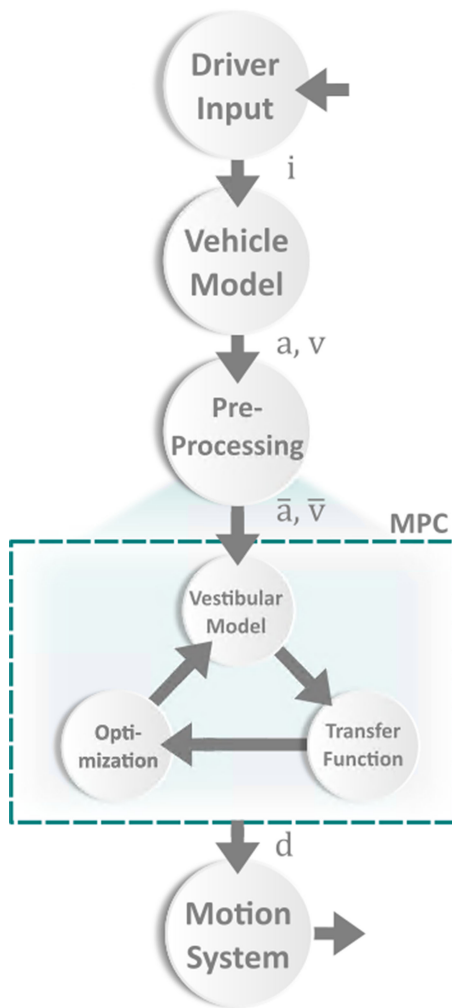


FIGURE 4. Diagram of an MPC motion cueing algorithm [42].

filter [53]. As shown in [54] and as depicted by the workflow in Fig. 4, the inputs (i) given by the real driver are received by the vehicle dynamics model which passes forward the vehicles acceleration and angular velocities (a, v). This information must then be pre-processed to build a predicted sequence of reference variables. Often, the next step is to assess the vestibular model to build the perceived values of acceleration and velocity. Other methods might simply use the angular velocity and specific force. As a final step, the system displacement (d) is calculated in order to track the perceived values before being handled to the motion control system. The importance of the reference over the MPC prediction horizon is highlighted in [55] where the future behavior of the driver is modelled as an optimal controller to have its behavior predicted and in [56] where the motion scale is treated as an optimization variable and a model of the kinematics of a 7-DOF motion system is used for motion prediction. In this work it is shown that the separation of the penalty on the motion gain from the overall target perception benefits the motion reproduction since excessive high-pass filtering

and tilt-coordination becomes not required when reproducing long-period forces.

Although the process in the diagram is simple, the computational burden is complex given the presence of so many complex models. In [57], both the human vestibular system and dynamic platform are modelled. For the vestibular system, linear transfer functions for the otoliths (translational motion) and for the semicircular canals (angular motion) are considered. A multi-objective problem is solved using the sorting genetic algorithm-II (NSGA-II) method, where the outputs are the gains for the translational and angular displacements and velocities of the platform. The proposed nonlinear MPC in [58] uses a multi-sensory cueing algorithm (MSCA) that accounts for the coordinates of the 9-DOF motion platform together with the coordinates for the active seat and active seatbelt. In addition, a model of a seated driver is added to the vestibular model to predict the effects of the forces generated by the seat and seatbelts. It is shown that this approach has a great impact in compact simulator since the use of space is optimized.

Although the MPC method has the potential to optimize the use of space and the driver experience, its often-understated limitation resides in how far ahead it can foresee without compromising the real time requirement. It is important to highlight how imprecise prediction can undermine the fidelity of motion. Some works focused on different methods aiming to reduce the optimization time. In [59], a continuous-time recurrent neural network (RNN) is used. In this work the authors model the dynamic platform kinematics and use a RNN to compute and apply the optimal trajectory of the platform online.

It is possible to conclude that the MPC method allows better use of the available hardware and automated tuning of the cues when compared to the previous methods, although its implementation requires complex modelling of different systems and massive computation power. Nevertheless, the MPC approach is considered promising for enabling state-of-the-art simulators to deliver best possible motion cueing.

G. KINEMATICS

The kinematics algorithm plays an important role connecting the motion cueing commands with the motion systems structure and hardware. The output of the motion cueing algorithm is trajectory of the cabin in Cartesian space, and this trajectory must be achieved by varying the length of the actuators. Therefore, actuators action is a function of desired cabin motion. As explained in [60] and [61], the concept of kinematics in driving simulation can be classified either as forward kinematics or inverse kinematics. The authors agree that forward kinematics is the process of determining the position of the motion platform once the actuators lengths are known (or predefined). Inverse kinematics is the process of acquiring the actuators length given the desired position of the motion platform. Likewise, in [62], inverse kinematics is defined in a more generic way as the process of calculating the joint coordinates from the end-effector coordinates. In

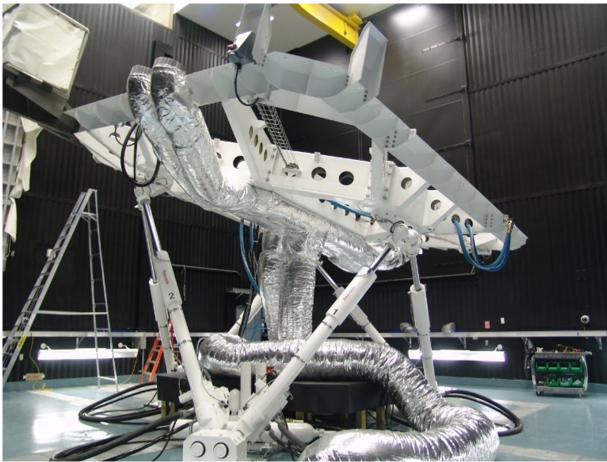


FIGURE 5. Hexapod motion system.

driving simulation, the end-effector is the mock-up cabin, and the joints are the couplings of each actuator to the motion platform. Since actuator length is a function of the desired motion of the cabin as previously mentioned, authors often specify the kinematics process in driving simulators as inverse kinematics.

The Stewart platform mechanism powered by electric/hydraulic actuators is very common among the state-of-the-art driving simulators. In [63], the authors describe a virtual modeling process of defining the kinematics of a Stewart platform and its boundaries, accounting for the type of the actuators, joints, and hinges. Stability is key to achieve real driving experience. Therefore, other studies focus on control methods to assure stability of those platforms. In [64], the requested position of the motion platform is performed using high order sliding mode control. In [65], a virtual PD+ controller on the basis of the passivity properties of the system was design to assure that the mechanism is asymptotically stable.

H. MOTION SYSTEM

The better the motion system is matched with the vehicle dynamics model, likely the higher the fidelity of the motion will be. There are several mechanisms to reproduce vehicle motion, from low fidelity systems with 2-DOF [66] and 3-DOF [67], to high fidelity systems with up to 13-DOF [68]. These mechanisms can be comprised of serial and/or parallel actuators. The parallel manipulators mechanism is a synergistic machine, where multiple actuators work together to support a single platform. The most common arrangement of parallel actuators applied in driving and flight simulators is the hexapod configuration, also called a Stewart platform, which is a compact design allowing for 6-DOF. The hexapod provides translational movement along, and rotational movement about, the three axes, x , y , and z [6]. As depicted in Fig. 5, the hexapod consists of a lower platform and an upper platform coupled together with six identical linear actuators. Another example of parallel actuators is the tripod, with one actuator at each corner of the moving platform it allows 3-DOF; roll,

pitch and heave [67]. One advantage of these type of systems over robotic manipulators is that an entire vehicle or a vehicle cabin can be mounted on them as seen in several cases. That represents a level of immersion difficult to achieve with a robotic arm manipulator for example.

In a serial mechanism, actuators are stacked and work independently, the most common example in industrial automation being a series of links connected by actuated joints from the base to one end-effector. An example of using this arrangement for simulator motion is depicted in [69] where a jointed robot arm moves around a motion envelope in order to create the motion cues. In the cited work, the motion trajectory was optimized for the 8-DOF motion system, combining the six-axis serial manipulator with a linear track and translational-actuated cabin. Although this has some advantages over the hexapod system, such as a larger motion envelope typically, it has some drawbacks. These include the limitations of speed and acceleration imposed by the joints, the lack of immersion given its form, a high degree of motion noise created by the high friction system, and the most notorious, the need for a very specific inverse kinematics algorithm.

Although many state-of-the-art simulators use a hexapod as the primary source of translational and rotational motion, the most advanced motion systems employ additional mechanisms creating redundant degrees of freedom for increased range of motion in a hybrid combination of parallel and serial actuation. Adding a turntable below the vehicle cabin or on top of the upper platform may help reduce simulator sickness as well as minimizing frame drops in the visual system by removing the need for the visuals to render a high yaw velocity, such investigation is conducted in [70]. Another hybrid system employed is coupling a linear XY motion structure to the lower platform, for a much larger longitudinal and lateral displacement. The large motion area set-up will provide most of the midrange frequency longitudinal and lateral motion, filling the large transfer function dip between the high-passed filtered specific forces and the low-pass filtered tilt coordination inherent with a hexapod alone. A sizing method for this type of configuration is presented in [71].

Some advanced driving simulators have this XY motion in the form of a rail system like in the University of Leeds Driving Simulator [72], the Stuttgart Driving Simulator located in Stuttgart University [73], others in the form of a steel belt drive on oil and bearings like the National Advanced Driving Simulator (NADS-1) [67] located in Iowa, United States. As explained in [68] the motion system of the NADS-1 presents 13-DOF achieved by the combination of a turntable (able to turn 330 degrees) mounted on top of a hexapod (6-DOF) that is attached to an XY platform with a large displacement area (close to 400 m). A dome is assembled on the upper part of the turntable to house the vehicle cabin that is mounted above high-frequency electric actuators. the XY rail track featured in the VTI IV simulator of the Swedish National Road Transport Research Institute [74]. In this work, the competence of this set-up (hexapod + rail track) in providing a motion that delivers realistic cues is proven. Experienced

drivers were capable of noticing detailed handling differences caused by variations of the vehicle parameters.

Although the system presented in [68] is considered to be one of the most capable of providing an accurate reproduction of real-world motion in a driving simulator, building systems such as this is not feasible for most testing centers, research institutes, or even vehicle manufactures due to the very high cost and physical space required. Using the concept of a hexapod and linear motion mechanisms, the Driver in Motion (DiM) 9-DOF platform was designed. As detailed in [75], this type of motion system (DiM 150 and DiM 250) relies on the hexapod to provide rotational motion as well as small linear displacements. In addition, the hexapod is coupled to a planar tripod frame that slides on a flat surface through air pads, which allows lateral, longitudinal and yaw motion. This redundant DOF design provides larger workspace use and system bandwidth when compared to the classical hexapods [54], but at a lower cost than larger linear motion systems. Other configuration of the DiM simulator is presented in [59], DiM 700 uses a central pulley (called discframe) driven by four cables. With this configuration, the DiM motion system achieves an even larger motion envelope, which makes it a more suitable choice for applications that require the reproduction of longer sustained accelerations.

The DiM solution is objective of study of several papers regarding MPC motion cueing algorithms such as [76], and figures as one product of higher motion fidelity. When building a simulator, the definition of the motion system is key, and should be done based on the intended application. The work in [77] develops an objective approach to be used when selecting between a 2-DOF, 3-DOF, or 6-DOF motion systems. The work also proposes a method to optimize the motion system geometry based on the future application of the driving simulator. Likewise, the work in [71] proposes an approach to lower the cost of the set-up by optimizing the size of the motion system accounting for the available workspace and a high-fidelity motion response. Both works can be used together as relevant guides for institutes looking for designing their own driving simulators. When designing a motion system, the actuators are equally as important as the architecture. Most recent simulator hexapods use electric actuators in place of hydraulics due to their lower cost and maintenance. They also allow for a higher motion bandwidth, but at a cost of higher friction and motion noise.

IV. STATE-OF-THE-ART DRIVING SIMULATORS

This session describes recent state-of-the-art driving simulators. The purpose is to give the reader examples of the application of previously mentioned systems. Older examples can be found in literature [9]. Although the following examples vary in number of degrees of freedom and fidelity, they all show a highly immersive environment provided by real vehicle mock-ups and large field of view. It is important to note the widespread use of desktop simulators in education and research centers, which are capable of meeting many research needs, but are not within the scope of the present survey.



FIGURE 6. MARCdrive.

A. MARCdrive

Static simulators are the most common structure applied in research center, both for the lower cost and simple implementation. Among this type of simulators is the MARCdrive lab installed in 2019 at the McMaster Automotive Resource Centre (MARC) in Hamilton, Canada, shown in Fig. 6. Although the MARCdrive is a static simulator, its high immersion is achieved by the real full vehicle facing a curved screen with 210 degrees of field of view (FoV) powered by three projectors. Low fidelity motion cues are given by the active seat and seatbelts. The first, built from air bladders inside the drivers seat, provides soft handling lateral cues (less than 10 Hz). The second pulls the drivers belt to provide longitudinal cues. Active steering and brakes also add realistic haptic cues. Different vehicle dynamics models such as CarSim [78] and VI-CarRealTime [79] can be used in the modular structure. The design of urban traffic scenarios is performed using Vires Virtual Test Drive VTD [80]. The main purpose of this driving simulator is for studies involving driver impairment, autonomous features, the analysis of hybrid powertrain, and energy management systems.

B. DriverLab

DriverLab, shown in Fig. 7 is Canada's most advanced driving simulator and part of the Challenging Environment Assessment Lab located at KITE, the research arm of the Toronto Rehabilitation Institute. DriverLab contains a full Audi A3 mounted on 7-DOF motion system consisting of a 360-degree turntable and a hydraulic hexapod. A curved 360-degree visual projection screen with 12 projectors rendered at 120 Hz and a surround sound system fully immerse the driver in the virtual world. To maximize the level of realism, DriverLab offers two novel features, a rain simulator that produces real water droplets on the windshield and a robotic glare simulator that recreates the harsh glare of oncoming headlights at night [33]. SCANerStudio [81] defines the scenario and



FIGURE 7. DriverLab.

renders the environment, while CarSim is used for the vehicle dynamics model. DriverLab is used to investigate driving safety in challenging conditions with various populations, including the effects of medications, brain injuries, sensory impairments, chronic pain, and neurodegenerative cognitive impairments.

C. UNIVERSITY OF LEEDS DRIVING SIMULATOR

The driving simulator of the University of Leeds is a worldwide reference in driving simulation ever since its launch in 2006. Its 4 m diameter dome houses a complete 2005 Jaguar S-type. The FoV is 250 degrees using 9 projectors and the virtual environment is rendered at a frequency of 60 Hz [82]. Its 8-DOF motion is achieved by the integration of a hexapod and an XY structure 5 m long [83]. The UoLDS also includes features such as haptic feedback through the pedals and steering wheel, and an eye tracking system to monitor driver behavior. Current research using the UoLDS includes driver monitoring, testing of automated driving systems, and road design.

D. FIAT CHRYSLER AUTOMOBILES VEHICLE DYNAMICS SIMULATOR

Inaugurated in 2019, at the Fiat Chrysler Automotive Research and Development Centre (ARDC), Windsor (Canada), the Vehicle Dynamics Simulator (VDC) showcases the next level of DiM technology. It features a DiM 250 driving simulator with a real vehicle cabin on top of a hexapod. The hexapod is coupled to a sliding table, with a tripod providing motion in the XY and yaw directions, which results in a 9-DOF

structure. Actuators are electric to provide high frequency responses. The visual system consists of three projectors with a 230 degree FoV screen. Vehicle dynamics is calculated by the VI-CarRealTime software, and graphics are provided by the VI-Graphics engine.

E. NATIONAL ADVANCED DRIVING SIMULATOR (NADS-1)

Although not new (1999), the NADS-1 simulator in Iowa is recognized to be one of the simulators with the highest level of fidelity in motion and immersion. With a combination of electric and hydraulic actuators and a 20 m $\times$ 20 m XY motion envelope, its 13-DOF motion system consists in the coupling of the hexapod and XY platform to a turntable where the dome sits on. The remaining 4-DOF are provided by high-frequency vibration actuators mounted replacing the front and rear suspension system at each wheel [68]. The dome receives projected image from 16 projectors resulting in 360 degrees of FoV. Research conducted at NADS-1 comprehends the field of driver-centered analysis as well as vehicle system-centered analysis.

V. APPLICATIONS

Driving simulation is broadly applied in research and industry. As discussed in [84], the use of driving simulators expands with the increasing necessity of developing fast, cost-effective, and safe means of testing interactivity between vehicle, passengers, and environment. The following section showcases some applications for different types of driving simulators.

A. DRIVER STUDIES

For all driver-in-the-loop simulators in the context of this review, the driver is an integral component of the total system being tested. However, one key branch of research is driver assessment, examining specifically the drivers behavior or performance rather than the vehicle systems. The most common use of driving simulation focused on the driver specifically are studies about driver training. Driving simulators can help drivers to enhance their ability and mitigate occurrence of accidents in the real world by presenting risky scenarios as in [85] and [86], typical common situations as in [87] and [88], or using both as shown in [89].

An example of testing driver performance is examining how substances such as alcohol and cannabis impair the ability to drive. To perform these experiments on real roads with real vehicles would be not only unethical but also irresponsible, while this kind of experiment can be safely performed in virtual environments. Examples using driving simulation with alcohol consumption is presented in [90] and [91]. In those studies, subjects were evaluated in low and high quantities of alcohol while others were provided with placebo. The authors agree that the ingestion of alcohol raises the risk-taking behavior of the drivers. They also agreed that lowering the legal limits of blood alcohol concentration (BAC) would lower the likelihood of accidents. Newest studies were similarly performed for cannabis in [92] and [93].

Simulation can also be used to assess sources of distractions. In [94] the influence surrounding vegetation has on drivers attention is evaluated through the use of an eye tracker. The results show that the vegetation plays a minor influence in longitudinal speed and lateral deviation although the roadside clear zone width can impact driving safety. Another use of an eye tracker system is presented in [95], where the system monitors drivers gaze to understand the effects of fatigue on eye tracking measures. It is shown that there is a significant change in the pupil when the driver is alert compared to a fatigued driver, demonstrating the effectiveness of the device on assessing fatigue. A similar device is introduced and tested in [96], using not only eye tracking but also heart rate and skin conductance measurements. Fatigue is also analyzed in [97], and how it leads to driver sleepiness during automated drives. In this paper a dedicated camera monitored head movement in addition to drivers gaze and the results showed that common features used for detecting drowsiness during manual driving might not be sufficient for performing the same detection during automated driving. Other driver assessments that would represent potential harm if performed in real environment but are easily done using simulators include testing mobile distractions [98], performance decay in older drivers [99], and the effects of daylight on sleepiness [100].

Driving simulators can also be used to test human machine interfaces (HMI). Some studies focus on how to design an interface that will be better for the driver to interact with, where simulators can be of great help. That is the case in [101], where the authors evaluate how drivers interact with a secondary device. The participants interacted with a touchscreen, rotary-controller, steering-wheel-controls, and a touchpad. This type of analysis can rate the ease of use of a device during the driving task and show what to employ depending on the purpose of the device. Again, performing these tests in real life could compromise the safety of the people involved. In addition, the ease of integration and testing of such systems in a driving simulator can be beneficial compared to implementing them in a real vehicle.

Most recently, the interaction between automated driving systems and drivers is being a subject of intense debate. In [102], the authors develop an approach to design and verify an HMI system to facilitate the transition from automated driving to manual driving. Other studies even propose interfaces that aim to build trust between passengers and machine in fully automated scenarios, such as in [103]. In this work the authors use VR system to provide the passenger a collection of cues that shares information from the sensors and the route planning.

B. AUTONOMOUS SYSTEMS DEVELOPMENT

Driving simulation can also be used to test autonomous and advanced driver assistance systems (ADAS). A very common driving assistant system is the adaptive cruise control (ACC). This feature allows the driver to specify its maximum speed and minimum distance to the preceding vehicle. Examples of this work can be seen in [104] where driving simulation is

used to explore the use of an ACC system in traffic situations, and in [105] where system failure is explored in different traffic scenarios.

One of the most important scenarios involving human interaction with ADAS is explored in [106], where the take-over maneuver is evaluated in different conditions. The volunteers had to face a forward collision situation with three different systems: autonomous vehicle with and without collision avoidance assistant, and manual drive mode. The results showed a lack of reaction by subjects when driving in the autonomous modes. In fact, the take-over maneuver is object of study of several papers regarding ADAS. In [107] take-over request is studied and an investigation of the orientation of the tactical alert given to the driver (whether towards to the hazard or away from it) is conducted. The transition is assessed in a lane change maneuver where the current lane presents a hazard four seconds ahead. In [108], tactile, visual, and auditory requests are given to the driver in order to understand the differences those modalities have in gaining drivers attention. The work in [109] highlights the importance of instruction and driver training on using ADAS.

Other works assessing take-over maneuvers can be found in [110] where different drivers are evaluated during a failure of the autonomous system, and in [111] where truck drivers face time critical take-over situations while performing a non-driving tasks. In [112], the authors evaluate how scheduled manual driving affects drowsiness and contribute to better take-over maneuvers when needed. In [113], an autonomous emergency braking system (AEB) that adapts to road friction is evaluated in comparison to a non-adaptable system. This scenario illustrates the highest level of assistance a driving simulator can provide during vehicle design because it allies the validation of a new safety feature in a driving condition very difficult to reproduce consistently (i.e. low-friction snowy road). It also highlights the importance of the ability to adapt to a changing situation or condition. Another example of AEB assessment is presented in [114], where several vehicle-bicycle imminent collision events are proposed with the objective of redesigning validation tests in order to cover a larger number of scenarios. A very well-known system is the lane keeping assistant. As described in [115], the lane keeping system helps the driver to keep the vehicle on a lane by applying torque (also called haptic feedback) to the steering wheel. In this work, the impact this system has on drivers fatigue during monotonous driving is investigated by monitoring the standard deviation of lateral position. The benefit of haptic feedback is also evaluated in [116], where the authors found that whole-body feedback during curves can be effective in avoiding hazardous situations when take-over is needed. Furthermore, in [117], a real-time adaptable haptic feedback is proposed based on the level of distraction of the driver.

Finally, driving simulation can be used to test the interaction between fully autonomous vehicles and passengers. In [118], the authors investigate the preference of passengers regarding the level of information the autonomous system

exchanges, such as sensor information, directions, and status. This study points out that different users will require different levels of information, which should be accounted for when designing autonomous systems. Smoothness of the autonomous driving is assessed in [72], where a non-linear model predictive control (NMPC) is developed to govern the vehicle trajectory, accounting for safety and comfort, intended to develop a system with a human-like behavior. Another assessment of an autonomous driving system from the perspective of the passengers is depicted in [119]. Here, passengers eye movement is tracked using eye tracking systems to rate the trust in the system. In this work, the results show that driving styles appeared to affect the trust. This shows a relevant paradigm in the use of driving simulators for autonomous systems testing. Not only should safety and efficiency be pursued, but also interaction and compliance with passengers expectations of motion, to achieve mainstream acceptability and a comfortable driving experience.

C. CHASSIS SYSTEMS DEVELOPMENT

In industry, one of the mainstream uses of driving simulators is for subjective dynamic evaluation and pre-design of chassis components. Although this is one of the larger use cases, due to the confidential nature of industry work, publications are not common. The work performed in [74] proves that experienced drivers are able to identify handling differences given parameter modifications using a driving simulator. The authors used different vehicle parameters such as roll stiffness and tire compliance to reproduced well-known maneuvers, e.g. single lane change. The driver subjective feeling was compared, as well as a comparison between the actual measurements and virtual values of yaw rate, roll, and steering torque. The results motivate works tuning lateral dynamics through suspension, steering, and active control systems. A similar test is conducted in [120], where acceleration of the drivers head was measured during a braking maneuver in a real vehicle and in a 6-DOF simulator. The study was to investigate if the whiplash motion of the neck could be reproduced in the simulator, validating its use for longitudinal dynamics assessments. In [121], different configurations of the steering system are tested by several subjects in three different scenarios, to evaluate how the environment and the system influence the drivers perception of controllability, ease of use and fun while driving. The work developed in [122] goes even deeper into an active steering system and proposed a new model that incorporates drivers gaze, justified by the fact that the steering behavior of drivers change with sightline distances. The results show that this concept can be further used in autonomous systems, since it was shown to achieve more precision than current proportional-integral-derivative (PID) controllers.

As mentioned before, suspension systems are also an object of study in driving simulation, due to the ease of changing components, configurations, geometry, and dynamics without prototyping parts. For active systems, different tunings can be assessed without the need for software compilation. Active

control systems such as electronic stability control (ESC) can also be tuned through driving simulation. The work detailed in [123] uses the driving simulator to prove the loss of control reduction provided by the ESC system. In that work, 120 subjects drove into typical crash situations with and without the assistance system. The results show the benefit of having ESC in all scenarios. Another work featuring an ESC system is depicted in [124], where vehicle behavior during anti-rollover interventions is investigated. This kind of test is important as it shows how safety systems behave in standard validation maneuvers compared to how well they perform in real world experiments.

More recent work described in [83] assesses the validity of a driving simulator in mimicking the dynamics of a low-friction test track, showing that although driving simulation can be used for dynamic assessment, there are still limitations in reproducing certain maneuvers and environments.

D. POWERTRAIN SYSTEMS DEVELOPMENT

Certain aspects of powertrain design can also be studied in driving simulation environment. Given the increase of pollution and imminent shortage of fuel resources, the electrification of powertrain systems is gaining momentum, with research in this field indicating that this technology will be key to solving the energy issue [125]. Several energy management strategies (EMS) have been developed in the past few years [126] for many different powertrain configurations [127] in consideration of the electrification trend, with the objective of reducing energy consumption and raising vehicle efficiency.

In addition, the importance of the journey mapping and driving conditions (e.g. pavement, weather, and traffic) is highlighted by several researchers [83] and [128]. Once more, driving simulation can be a powerful tool, reducing design timing and cost, providing controlled environments, and allowing integration of technologies beforehand. In powertrain design, the HMI analysis is also important. An investigation on how augmented reality (AR) displays can guide the driver during the driving task is proposed in [129], where a hybrid electric vehicle (HEV) is emulated using a co-simulation between a hardware in the loop (HIL) test rig and a driving simulator. This study showed that fuel efficiency can be increased using simple display systems that guide the driver during the journey. The same concept is explored in [130] and [131], where the ability to foresee events using a V2X (vehicle to road/vehicle/database communication) is used in order to display instructions to the driver.

Using the same concept of driver guidance, several eco-driving studies were performed in [132] and [133] to evaluate fuel consumption improvement by instructing drivers on eco-friendly driving behavior. The work in [134] investigates the role gas pedal feedback can play in making drivers achieve a higher fuel efficiency. In this work two different pedal systems were tested by twenty drivers. The results show that the most fuel-efficient behavior is achieved when drivers are provided

guiding force feedback (such as a step change in the pedal force) compared to a system that provided increased firmness.

In another study examining fuel consumption, [135] integrates a static simulator with a dynamometer. This integration makes it possible to evaluate the performance of electric vehicle (EV) systems in tests performed by real human in several driving conditions. In [136], ten different drivers were placed in a car following scenario that replicates a realistic driver model for fuel economy assessment of a HEV. The different driver behaviors were analyzed and categorized. This data is used to develop a car following driver model for evaluating fuel consumption, and to create a reference virtual driver that can provide real time feedback to human drivers.

Subjective longitudinal acceleration can also be validated using driving simulation. The authors in [137] used a 6-DOF driving simulator to present different engine configurations to drivers in a tip-in maneuver defined by a sudden step on the accelerator pedal from a constant engine speed (1500 rpm in second gear) without gearshift. The results showed that drivers have the sensibility to perceive changes in the acceleration profile for different engine configurations. That indicates that driving simulators can be used for subjective drivability assessment of powertrains prior to prototyping. In addition, the authors in [138] have shown that it is possible to categorize acceleration levels. That is an important insight to HEV developers since drivability and drivers perception has very tight requirements.

E. ROAD DESIGN

In addition to the aforementioned vehicle design, road design can also be studied and improved through driving simulation. In [139], driving performance is evaluated for different configurations of road signs. A similar work is presented in [140], where audio warnings through differing pavement surfaces (rumble strips) in addition to visual warnings were examined, something that would be costly to create in a physical environment.

VI. CONCLUSION

This review has explained the mechanisms of driving simulation technology by describing each system in the architecture. The human perceptual system was also briefly explained since the purpose of a driving simulator is to evoke a desired response from that system while imitating the driving experience.

Current state-of-the-art simulators are also presented, together with their most important characteristics and typical uses. These state-of-the-art systems aim for the highest level of fidelity by providing highly immersive set-ups with full vehicle mock-ups, a large field of view, and accurate vehicle dynamics models. The ability to create and control driving conditions and scenarios such as weather and traffic are also key. When using motion, these simulators also include well-developed motion cueing algorithms that account not only for the limitations of the human perceptual system but also for the boundaries of the motion system. Static simulators can often

be used in studies where the focus is not primarily closed loop control of the vehicle by a human driver, while motion simulators offer an increased scope of use, including analysis of chassis control and autonomous features.

The final section has shown several examples of driving simulation use. It is important to highlight that the three of the most important benefits of driving simulation are the safety for subject drivers and other road users, the reduction of time in design phase, and the ability to reproduce a variety of controlled scenarios and conditions. In addition, it is possible to foresee positive prospects for expanded driving simulation use. The increasing momentum of autonomous and active control systems means that more virtual validation of these systems will be required in the future. Further, despite some claims that autonomous driving would mean less driver-in-the-loop simulation (due to less human driving), the need for a platform to test system acceptability and to train users will require increased use of simulation, similar to patterns that can be observed in the history of aerospace development. The need for environments that show no harm for the occupants and that can produce precise and reproducible emergency situations will also increase. The same increased use of simulation also holds for the electrification process.

Although some studies were already performed assessing powertrain systems, these analyses will become more common and precise as the need to virtually validate those systems increases. Moreover, driving simulation allows the validation of the interaction between different systems, both at the component and system level. It enables engineers to test the integration between autonomous systems with active safety systems for example, as well as analyzing it from the drivers perspective. This integrated interactive approach is key to develop systems that work as one.

Finally, the next generation of driving simulators will show high connectivity with other testing apparatuses, such as human condition tracking devices, mobile phones, and dynamometers. Although few works were found in this area, the necessity of integrating testing systems is clear given the necessity of integrating vehicle systems. The use of driving simulation is destined to increase, saving time as well as raising the safety of tests.

REFERENCES

- [1] J. Slob, "State-of-the-art driving simulators, a literature survey," Eindhoven Univ. Technol., Eindhoven, Netherlands, Tech. Rep. DCT 2008.107, Aug. 2008.
- [2] H. H. Valverde, "A review of flight simulator transfer of training studies," *Hum. Factors*, vol. 15, no. 6, pp. 510–522, 1973.
- [3] G. Miles and D. Vincent, "The institutes tests for motor drivers," *The Hum. Factor*, vol. 8, no. 7, pp. 245–257, 1934.
- [4] W. W. Wierwille and P. P. Fung, "Comparison of computer-generated and simulated motion picture displays in a driving simulation," *Hum. Factors*, vol. 17, no. 6, pp. 577–590, 1975.
- [5] J. Gruening, J. Bernard, C. Clover, and K. Hoffmeister, "Driving simulation," *SAE Trans.*, vol. 107, pp. 376–385, 1998.
- [6] E. Blana, A survey of driving research simulators around the world. Institute for Transport Studies, University of Leeds, Leeds, U.K., Dec. 1996. [Online]. Available: <http://eprints.whiterose.ac.uk/2110/>

- [7] D. L. Fisher, M. Rizzo, J. Caird, and J. D. Lee, *Handbook of Driving Simulation for Engineering, Medicine, and Psychology*. Boca Raton, FL: CRC Press, 2011.
- [8] A. H. J. Jamson, "Motion cueing in driving simulators for research applications," Ph.D. dissertation, Univ. Leeds, Leeds, U.K., Nov. 2010. [Online]. Available: <http://etheses.whiterose.ac.uk/1915/>
- [9] N. Mohajer, H. Abdi, K. Nelson, and S. Nahavandi, "Vehicle motion simulators, a key step towards road vehicle dynamics improvement," *Veh. Syst. Dyn.*, vol. 53, no. 8, pp. 1204–1226, 2015.
- [10] P. Zaai, F. Nieuwenhuizen, M. Mulder, and M. van Paassen, "Perception of visual and motion cues during control of self-motion in optic flow environments," presented at the AIAA Model. Simul. Technol. Conf. Exhib., Keystone, CO, USA, Aug. 21–24, 2006, Paper 6627.
- [11] R. Wilkie and J. Wann, "Controlling steering and judging heading: Retinal flow, visual direction, and extraretinal information," *J. Exp. Psychol.: Hum. Perception Perform.*, vol. 29, no. 2, pp. 363–378, 2003.
- [12] B. Keshavarz, J. L. Campos, P. R. DeLucia, and D. Oberfeld, "Estimating the relative weights of visual and auditory tau versus heuristic-based cues for time-to-contact judgments in realistic, familiar scenes by older and younger adults," *Attention, Perception, Psychophys.*, vol. 79, no. 3, pp. 929–944, 2017.
- [13] B. Keshavarz, L. J. Hettlinger, D. Vena, and J. L. Campos, "Combined effects of auditory and visual cues on the perception of vection," *Exp. Brain Res.*, vol. 232, no. 3, pp. 827–836, 2014.
- [14] J. R. Lackner and P. DiZio, "Vestibular, proprioceptive, and haptic contributions to spatial orientation," *Annu. Rev. Psychol.*, vol. 56, pp. 115–147, 2005.
- [15] E. P. Widmaier, H. Raff, K. T. Strang, and A. J. Vander, *Vander's Hum. Physiology: The mechanisms of body function*. Boston, MA: McGraw-Hill Higher Education, 2008.
- [16] G. Markkula, R. Romano, R. Waldram, O. Giles, C. Mole, and R. Wilkie, "Modelling visual-vestibular integration and behavioural adaptation in the driving simulator," *Transp. Res. Part F: Traffic Psychol. Behav.*, vol. 66, pp. 310–323, 2019.
- [17] Smart: Servier medical art - ears. Accessed: Sep. 7, 2020. [Online]. Available: <https://smart.servier.com/category/anatomy-and-the-human-body/auditory-system/ears/>
- [18] O. Balci, "Verification, validation, and certification of modeling and simulation applications," in *Proc. Winter Simul. Conf.*, 2003, vol. 1, pp. 150–158.
- [19] R. N. Jazar, *Vehicle Dynamics: Theory and Application*. Cham, Switzerland: Springer, 2017.
- [20] D. Schramm, M. Hiller, and R. Bardini, *Vehicle Dynamics: Modeling and Simulation*. Berlin, Germany: Springer, 2014.
- [21] T. D. Gillespie, *Fundamentals of Vehicle Dynamics*. Warrendale, PA, USA: Society of Automotive Engineers, 1992.
- [22] M. Matowicki, O. Přibyl, and P. Bouchner, "Pragmatic overview of surrounding traffic implementation into driving simulator," in *Proc. ELEKTRO*, 2016, pp. 423–428.
- [23] J. Sun, Z. Ma, T. Li, and D. Niu, "Development and application of an integrated traffic simulation and multi-driving simulators," *Simul. Modelling Pract. Theory*, vol. 59, pp. 1–17, 2015.
- [24] L. Yang, X. Li, W. Guan, H. M. Zhang, and L. Fan, "Effect of traffic density on drivers lane change and overtaking maneuvers in free-way situation driving simulator based study," *Traffic Inj. Prevention*, vol. 19, no. 6, pp. 594–600, 2018.
- [25] S. Sharples, S. Shalloe, G. Burnett, and D. Crundall, "Journey decision making: The influence on drivers of dynamic information presented on variable message signs," *Cognit., Technol. Work*, vol. 18, no. 2, pp. 303–317, 2016.
- [26] D. Xiao, Y. Fang, Y. Zhang, and Z. Guo, "Analysis of driving behavior at the bridge-tunnel transition section in reduced visibility situations," in *Proc. 4th Int. Conf. Transp. Inf. Safety*, 2017, pp. 581–588.
- [27] X. Chang, H. Li, L. Qin, J. Rong, Y. Lu, and X. Chen, "Evaluation of cooperative systems on driver behavior in heavy fog condition based on a driving simulator," *Accident Anal. Prevention*, vol. 128, pp. 197–205, 2019.
- [28] R. Kemmerer Jr, S. Hulbert, and R. Donohue, "UCLA driving simulation laboratory: With a 360 degree scene around a full size car," in *Proc. Simulators Simul. II: Des., Appl. Technol.*, Int. Soc. Optics Photon., 1975, vol. 59, pp. 158–170.
- [29] B. Blissung and F. Bruzelius, "Exploring the suitability of virtual reality for driving simulation," in *Proc. Driving Simul. Conf.*, 2018, pp. 163–166.
- [30] C. Zller, A. Miller, L. Eggert, H. Winner, and B. Abendroth, "Applicability of head-mounted displays in driving simulation," in *Proc. Driving Simul. Conf. Eur. VR*, A. Kemeny, F. Colombet, F. Merienne, and S. Espi, Eds., Strasbourg, France: Driving Simulation Association, 2019, pp. 9–15.
- [31] F. Colombet, D. Paillot, F. Mérienne, and A. Kemeny, "Visual scale factor for speed perception," *J. Comput. Inf. Sci. Eng.*, vol. 11, no. 4, 2011.
- [32] H. Schmieder, K. Nagel, and H.-P. Schoener, "Enhancing a driving simulator with a 3d-stereo projection system," in *DSC 2017 Eur. Driving Simulation Association*, 2017, pp. 103–110.
- [33] B. Haycock, J. Campos, N. Koenraad, M. Potter, and S. Advani, "Creating headlight glare in a driving simulator," *Transp. Res. Part F: Traffic Psychol. Behav.*, vol. 61, pp. 93–106, 2019.
- [34] The openscene graph project. Accessed: Sep. 1, 2020. [Online]. Available: <http://www.openscenegraph.org/>
- [35] Unity for all. Accessed: Sep. 1, 2020. [Online]. Available: <https://unity.com/>
- [36] Unreal engine. make something unreal. Accessed: Sep. 1, 2020. [Online]. Available: <https://www.unrealengine.com/en-US/>
- [37] C. Merenda, C. Suga, J. Gabbard, and T. Misu, "Effects of vehicle simulation visual fidelity on assessing driver performance and behavior," in *Proc. IEEE Intell. Veh. Symp. (IV)*, 2019, pp. 1679–1686.
- [38] J. Leudet, F. Christophe, T. Mikkonen, and T. Männistö, "Ailivesim: An extensible virtual environment for training autonomous vehicles," in *Proc. IEEE 43rd Annu. Comput. Softw. Appl. Conf. (COMPSAC)*, vol. 1, 2019, pp. 479–488.
- [39] D. A. Heitbrink and S. Cable, "Design of a driving simulation sound engine," presented at the Driving Simul. Conf., North Amer. 2007 (DSC-NA 2007), Iowa City, IA, USA, Sep. 12–14, 2007, Paper N2007-028.
- [40] R. Ramkhalawansingh, B. Keshavarz, B. Haycock, S. Shahab, and J. L. Campos, "Age differences in visual-auditory self-motion perception during a simulated driving task," *Front. Psychol.*, vol. 7, pp. 595.1–595.12, 2016. [Online]. Available: <https://app.dimensions.ai/details/publication/pub.1014944210>
- [41] R. Suikat, "The new dynamic driving simulator at dlr," in *Proc. Driving Simulator Conf.*, no. 8, 2005, pp. 374–381.
- [42] M. Bruschetta, F. Maran, and A. Beghi, "A fast implementation of mpc-based motion cueing algorithms for mid-size road vehicle motion simulators," *Veh. Syst. Dyn.*, vol. 55, no. 6, pp. 802–826, 2017.
- [43] L. Nehaoua, H. Mohellebi, A. Amouri, H. Arioui, S. Espie, and A. Kheddar, "Design and control of a small-clearance driving simulator," *IEEE Trans. Veh. Technol.*, vol. 57, no. 2, pp. 736–746, Mar. 2008.
- [44] Z. Fang, M. Tsushima, E. Kitahara, N. Machida, D. Wautier, and A. Kemeny, "Motion cueing algorithm for high performance driving simulator using yaw table," *IFAC-PapersOnLine*, vol. 50, no. 1, pp. 15 965–15 970, 2017.
- [45] L. Reid and M. Nahon, "Flight simulation motion-base drive algorithms: Part 1. Developing and testing equations," UTIAS Rep., no. 296, 1985.
- [46] Z. Fang and A. Kemeny, "Motion cueing algorithms for a real-time automobile driving simulator," in *Proc. Driving Simul. Conf.*, 2012, pp. 159–174.
- [47] H. Asadi, C. P. Lim, S. Mohamed, D. Nahavandi, and S. Nahavandi, "Increasing motion fidelity in driving simulators using a fuzzy-based washout filter," *IEEE Trans. Intell. Veh.*, vol. 4, no. 2, pp. 298–308, Jun. 2019.
- [48] T. Miunske, C. Holzapfel, E. Baumgartner, and H.-C. Reuss, "A new approach for an adaptive linear quadratic regulated motion cueing algorithm for an 8 DoF full motion driving simulator," in *Proc. Int. Conf. Robot. Autom.*, 2019, pp. 497–503.
- [49] S.-H. Chen and L.-C. Fu, "An optimal washout filter design for a motion platform with senseless and angular scaling maneuvers," in *Proc. Amer. Control Conf.*, 2010, pp. 4295–4300.
- [50] H. Asadi, S. Mohamed, C. P. Lim, and S. Nahavandi, "Robust optimal motion cueing algorithm based on the linear quadratic regulator method and a genetic algorithm," *IEEE Trans. Syst., Man, Cybern.: Syst.*, vol. 47, no. 2, pp. 238–254, Feb. 2017.
- [51] R. Sivan, J. Ish-Shalom, and J.-K. Huang, "An optimal control approach to the design of moving flight simulators," *IEEE Trans. Syst., Man, Cybern.*, vol. 12, no. 6, pp. 818–827, Nov. 1982.

- [52] R. Telban, F. Cardullo, and J. Houck, "A nonlinear, human-centered approach to motion cueing with a neurocomputing solver," presented at the AIAA Model. Simul. Technol. Conf. Exhib., Monterey, CA, USA, Aug. 5–8, 2002, Paper 4692.
- [53] A. I. Aulia, H. Hindersah, A. S. Rohman, and E. Hidayat, "Design of MPC-based motion cueing for 4 DoF simulator platform," in *Proc. IEEE 9th Int. Conf. Syst. Eng. Technol.*, 2019, pp. 183–188.
- [54] M. Bruschetta, C. Cenedese, A. Beghi, and F. Maran, "A motion cueing algorithm with look-ahead and driver characterization: Application to vertical car dynamics," *IEEE Trans. Hum.-Mach. Syst.*, vol. 48, no. 1, pp. 6–16, Feb. 2018.
- [55] A. Lamprecht, D. Steffen, J. Haecker, and K. Graichen, "Optimal control based reference generation for model predictive motion cueing algorithms," in *Proc. IEEE Conf. Control Technol. Appl.*, 2019, pp. 203–208.
- [56] A. Bukal, B. Haycock, and P. R. Grant, "An adaptive model predictive control based motion drive algorithm," presented at the AIAA Scitech. 2019 Forum, San Diego, CA, USA, Jan. 7–11, 2019, Paper 0423.
- [57] A. Mohammadi, H. Asadi, S. Mohamed, K. Nelson, and S. Nahavandi, "Multiobjective and interactive genetic algorithms for weight tuning of a model predictive control-based motion cueing algorithm," *IEEE Trans. Cybern.*, vol. 49, no. 9, pp. 3471–3481, 2018.
- [58] M. Bruschetta, Y. Chen, D. Cunico, E. Mion, and A. Beghi, "A nonlinear MPC based motion cueing strategy for a high performance driving simulator with active seat," in *Proc. IEEE 15th Int. Workshop Adv. Motion Control*, 2018, pp. 23–28.
- [59] C. RENGIFO, J.-R. Chardonnet, D. Paillot, H. Mohellebi, and A. Kemeny, "Solving the constrained problem in model predictive control based motion cueing algorithm with a neural network approach," in *Proc. Driving Simul. Conf. Eur. VR*, Antibes, France, Sep. 2018, pp. 63–69. [Online]. Available: <https://hal.archives-ouvertes.fr/hal-01940772>
- [60] Y. Chiew, M. A. Jalil, and M. Hussein, "Kinematic modeling of driving simulator motion platform," in *Proc. IEEE Conf. Innov. Technol. Intell. Syst. Ind. Appl.*, 2008, pp. 30–34.
- [61] J. G. Zeng, J. Y. Liu, and Q. Yu, "Design and development of 6dof hydraulic motion platform for vehicle driving simulator," in *Proc. Appl. Mech. Mater.*, vol. 505 Trans Tech Publ, 2014, pp. 315–318.
- [62] Y. Jin, H. Chanal, and F. Paccot, *Parallel Robots. Handbook of Manufacturing Engineering and Technology*. Dordrecht, The Netherlands: Springer, 2015.
- [63] G. Jia, G. Pan, Q. Gao, and Y. Zhang, "Research on position inverse solution of electric-driven stewart platform based on simulink," *J. Eng.*, vol. 2019, no. 13, pp. 379–383, 2019.
- [64] P. R. Kumar and B. Bandyopadhyay, "Stabilization of Stewart platform using higher order sliding mode control," in *Proc. 7th Int. Conf. Elect. Comput. Eng.*, 2012, pp. 945–948.
- [65] T. S. Tamir, G. Xiong, Y. Tian, and G. Xiong, "Passivity based control of stewart platform for trajectory tracking," in *Proc. 14th IEEE Conf. Ind. Electron. Appl.*, 2019, pp. 988–993.
- [66] C. Zhang and L. Zhang, "Kinematics analysis and workspace investigation of a novel 2-DoF parallel manipulator applied in vehicle driving simulator," *Robot. Comput.-Integr. Manuf.*, vol. 29, no. 4, pp. 113–120, 2013.
- [67] J. Suh, K. Yi, J. Jung, K. Lee, H. Chong, and B. Ko, "Design and evaluation of a model predictive vehicle control algorithm for automated driving using a vehicle traffic simulator," *Control Eng. Pract.*, vol. 51, pp. 92–107, 2016.
- [68] M. Wilkinson, T. Brown, and O. Ahmad, "The national advanced driving simulator (NADS) description and capabilities in vision-related research," *Optometry-J. Amer. Optometric Assoc.*, vol. 83, no. 6, pp. 285–288, 2012.
- [69] M. Katliar, M. Olivari, F. M. Drop, S. Nooij, M. Diehl, and H. H. Bülthoff, "Offline motion simulation framework: Optimizing motion simulator trajectories and parameters," *Transp. Res. part F: Traffic Psychol. Behav.*, vol. 66, pp. 29–46, 2019.
- [70] R. R. Maurant and Z. Yin, "A turning cabin simulator to reduce simulator sickness," in *Proc. Eng. Reality Virtual Reality, Int. Soc. Optics Photon.*, 2010, p. 752503.
- [71] E. Sadraei, R. Romano, S. Jamson, G. Markkula, and H. Jamson, "Driving simulator motion base right sizing," in *DSC 2018 Europe*. Driving Simulation Association, 2018, pp. 113–120.
- [72] C. Wei et al., "Risk-based autonomous vehicle motion control with considering human drivers behaviour," *Transp. Res. Part C: Emerg. Technol.*, vol. 107, pp. 1–14, 2019.
- [73] J. Pitz, M.-T. Nguyen, G. Baumann, and H.-C. Reuss, "Combined motion of a hexapod with a xy table system for lateral movements," in *DSC 2014 Europe*. Driving Simulation Association, 2014, pp. 06.1–06.2.
- [74] S. Kharrazi, B. Augusto, and N. Fröjd, "Vehicle dynamics testing in motion based driving simulators," *Veh. Syst. Dyn.*, vol. 58, no. 1, pp. 92–107, 2020.
- [75] M. Bruschetta, F. Maran, and A. Beghi, "A nonlinear, MPC-based motion cueing algorithm for a high-performance, nine-DoF dynamic simulator platform," *IEEE Trans. Control Syst. Technol.*, vol. 25, no. 2, pp. 686–694, Mar. 2017.
- [76] M. Bruschetta, C. Cenedese, and A. Beghi, "A real-time, MPC-based motion cueing algorithm with look-ahead and driver characterization," *Transp. Res. Part F: Traffic Psychol. Behav.*, vol. 61, pp. 38–52, 2019.
- [77] L. Bruck, S. Veldhuis, and A. Emadi, "Selection method of a driving simulator motion system," in *Proc. IEEE Transp. Electrification Conf. Expo.*, 2019, pp. 1–6.
- [78] Carsim mechanical simulation. Accessed: Sep. 1, 2020. [Online]. Available: <https://www.carsim.com/products/carsim/index.php>
- [79] Vi-carrealtime: One vehicle model from concept to sign-off. Accessed: Sep. 1, 2020. [Online]. Available: <https://www.vi-grade.com/en/products/vi-carrealtime/>
- [80] Virtual test drive: Enabling safety validation in autonomous driving and adas system simulation. Accessed: Sep. 1, 2020. [Online]. Available: <https://vires.msccsoftware.com/>
- [81] Scanner studio. Accessed: Sep. 1, 2020. [Online]. Available: <https://www.avsimulation.com/scannerstudio/>
- [82] E. Paschalidis, C. F. Choudhury, and S. Hess, "Combining driving simulator and physiological sensor data in a latent variable model to incorporate the effect of stress in car-following behaviour," *Anal. Methods Accident Res.*, vol. 22, p. 100089, 2019.
- [83] R. Romano et al., "An objective assessment of the utility of a driving simulator for low MU testing," *Transp. Res. Part F: Traffic Psychol. Behav.*, vol. 65, pp. 34–45, 2019.
- [84] L. Bruck, A. Emadi, and K. P. Divakarla, "A review of the relevance of driving condition mapping and vehicle simulation for energy management system design," *Int. J. Powertrains*, vol. 8, no. 3, pp. 224–251, 2019.
- [85] Y.-J. Ou, X.-L. Wang, J.-F. Jiang, H.-Y. Wei, C.-L. Huang, and K.-S. Hsu, "Simulator training to drive the risk perception of the reliability and validity," in *Proc. IEEE Int. Conf. Appl. Syst. Invention*, 2018, pp. 374–377.
- [86] J. H. Uurlings, E. Roelofs, A. Cuenen, K. Brijs, T. Brijs, and E. M. Jongen, "Development of single-session driving simulator-based and computer-based training for at-risk older drivers," *Educ. Gerontol.*, vol. 45, no. 4, pp. 283–296, 2019.
- [87] A. Ali, A. Elnaggar, D. Reichardt, and S. Abdennadher, "Gamified virtual reality driving simulator for asserting driving behaviors," in *Proc. 1st Int. Conf. Game, Game Art, Gamification.*, 2016, pp. 1–6.
- [88] T. Imamura, T. Ogi, E. T. C. Lun, Z. Zhang, and T. Miyake, "Trial study of traffic safety education for high school students using driving simulator," in *Proc. IEEE Int. Conf. Syst., Man, Cybern.*, 2013, pp. 4606–4611.
- [89] P. M. van Leeuwen, R. Happee, and J. C. de Winter, "Changes of driving performance and gaze behavior of novice drivers during a 30-min simulator-based training," *Procedia Manuf.*, vol. 3, pp. 3325–3332, 2015.
- [90] C. Irwin, E. Iudakhina, B. Desbrow, and D. McCartney, "Effects of acute alcohol consumption on measures of simulated driving: A systematic review and meta-analysis," *Accident Anal. Prevention*, vol. 102, pp. 248–266, 2017.
- [91] N. A. Van Dyke and M. T. Fillmore, "Laboratory analysis of risky driving at 0.05% and 0.08% blood alcohol concentration," *Drug Alcohol Dependence*, vol. 175, pp. 127–132, 2017.
- [92] A. Tank et al., "On the impact of cannabis consumption on traffic safety: A driving simulator study with habitual cannabis consumers," *Int. J. Legal Med.*, vol. 133, no. 5, pp. 1411–1420, 2019.
- [93] S. Hartley et al., "Effect of smoked cannabis on vigilance and accident risk using simulated driving in occasional and chronic users and the Pharmacokinetic-Pharmacodynamic relationship," *Clin. Chem.*, vol. 65, no. 5, pp. 684–693, 2019.

- [94] C. D. Fitzpatrick, S. Samuel, and M. A. Knodler Jr, "Evaluating the effect of vegetation and clear zone width on driver behavior using a driving simulator," *Transp. Res. Part F: Traffic Psychol. Behav.*, vol. 42, pp. 80–89, 2016.
- [95] J. Xu, J. Min, and J. Hu, "Real-time eye tracking for the assessment of driver fatigue," *Healthcare Technol. Lett.*, vol. 5, no. 2, pp. 54–58, 2018.
- [96] E. Schmidt and A. C. Bullinger, "Mitigating passive fatigue during monotonous drives with thermal stimuli: Insights into the effect of different stimulation durations," *Accident Anal. Prevention*, vol. 126, pp. 115–121, 2019.
- [97] J. Schmidt, C. Braunagel, W. Stolzmann, and K. Karrer-Gauß, "Driver drowsiness and behavior detection in prolonged conditionally automated drives," in *Proc. IEEE Intell. Veh. Symp. (IV)*, 2016, pp. 400–405.
- [98] K. P. Wijayarathna, M. L. Cunningham, M. A. Regan, S. Jian, S. Chand, and V. V. Dixit, "Mobile phone conversation distraction: Understanding differences in impact between simulator and naturalistic driving studies," *Accident Anal. Prevention*, vol. 129, pp. 108–118, 2019.
- [99] M. Kunishige, H. Fukuda, T. Iida, N. Kawabata, C. Ishizuki, and H. Miyaguchi, "Spatial navigation ability and gaze switching in older drivers: A driving simulator study," *Hong Kong J. Occup. Ther.*, vol. 32, no. 1, pp. 22–31, 2019.
- [100] C. Ahlström, A. Anund, C. Fors, and T. Åkerstedt, "The effect of daylight versus darkness on driver sleepiness: a driving simulator study," *J. Sleep Res.*, vol. 27, no. 3, p. e12642, 2018.
- [101] D. R. Large, G. Burnett, E. Crundall, G. Lawson, L. Skrypchuk, and A. Mouzakitis, "Evaluating secondary input devices to support an automotive touchscreen HMI: A cross-cultural simulator study conducted in the UK and China," *Appl. Ergonom.*, vol. 78, pp. 184–196, 2019.
- [102] S. Yun, T. Teshima, and H. Nishimura, "Hum.-machine interface design and verification for an automated driving system using system model and driving simulator," *IEEE Consum. Electron. Mag.*, vol. 8, no. 5, pp. 92–98, 2019.
- [103] L. Morra, F. Lamberti, F. G. Pratticó, S. La Rosa, and P. Montuschi, "Building trust in autonomous vehicles: Role of virtual reality driving simulators in hmi design," *IEEE Trans. Veh. Technol.*, vol. 68, no. 10, pp. 9438–9450, Oct. 2019.
- [104] E. Bénédicto and A. Dòria-Cerezo, "Influence of cooperative-controlled driving in the traffic flow," in *Proc. IEEE Int. Conf. Ind. Technol.*, 2018, pp. 1795–1800.
- [105] G. Bianchi Piccinini *et al.*, "How do drivers respond to silent automation failures? Driving simulator study and comparison of computational driver braking models," *Hum. Factors*, vol. 62, no. 7, pp. 1212–1229, 2020.
- [106] M. Blommer, R. Curry, R. Swaminathan, L. Tijerina, W. Talamonti, and D. Kochhar, "Driver brake vs. steer response to sudden forward collision scenario in manual and automated driving modes," *Transp. Res. Part F: Traffic Psychol. Behav.*, vol. 45, pp. 93–101, 2017.
- [107] G. Cohen-Lazry, N. Katzman, A. Borowsky, and T. Oron-Gilad, "Directional tactile alerts for take-over requests in highly-automated driving," *Transp. Res. Part F: Traffic Psychol. Behav.*, vol. 65, pp. 217–226, 2019.
- [108] S. Petermeijer, F. Doubek, and J. de Winter, "Driver response times to auditory, visual, and tactile take-over requests: A simulator study with 101 participants," in *Proc. IEEE Int. Conf. Syst., Man, Cybern.*, 2017, pp. 1505–1510.
- [109] T. Ogitsu and H. Mizoguchi, "A study on driver training on advanced driver assistance systems by using a driving simulator," in *Proc. Int. Conf. Connected Veh. Expo.*, 2015, pp. 352–353.
- [110] F. M. Favaro, P. Seewald, M. Scholtes, and S. Eurich, "Quality of control takeover following disengagements in semi-automated vehicles," *Transp. Res. Part F: Traffic Psychol. Behav.*, vol. 64, pp. 196–212, 2019.
- [111] A. Lotz, N. Russwinkel, and E. Wohlfarth, "Response times and gaze behavior of truck drivers in time critical conditional automated driving take-overs," *Transp. Res. Part F: Traffic Psychol. Behav.*, vol. 64, pp. 532–551, 2019.
- [112] Y. Wu, K. Kihara, Y. Takeda, T. Sato, M. Akamatsu, and S. Kitazaki, "Effects of scheduled manual driving on drowsiness and response to take over request: A simulator study towards understanding drivers in automated driving," *Accident Anal. Prevention*, vol. 124, pp. 202–209, 2019.
- [113] I. Koglbauer, J. Holzinger, A. Eichberger, and C. Lex, "Autonomous emergency braking systems adapted to snowy road conditions improve drivers' perceived safety and trust," *Traffic Inj. Prevention*, vol. 19, no. 3, pp. 332–337, 2018.
- [114] J. Duan *et al.*, "Driver braking behavior analysis to improve autonomous emergency braking systems in typical chinese vehicle-bicycle conflicts," *Accident Anal. Prevention*, vol. 108, pp. 74–82, 2017.
- [115] Z. Wang, R. Zheng, T. Kaizuka, K. Shimono, and K. Nakano, "The effect of a haptic guidance steering system on fatigue-related driver behavior," *IEEE Trans. Hum.-Mach. Syst.*, vol. 47, no. 5, pp. 741–748, Oct. 2017.
- [116] E. Pakdamanian, L. Feng, and I. Kim, "The effect of whole-body haptic feedback on drivers perception in negotiating a curve," *Proc. Hum. Factors Ergonom. Soc. Annu. Meet.*, Los Angeles, CA, vol. 62, no. 1, 2018, pp. 19–23.
- [117] M. Benloucif, C. Sentouh, J. Floris, P. Simon, and J.-C. Popieul, "On-line adaptation of the level of haptic authority in a lane keeping system considering the drivers state," *Transp. Res. Part F: Traffic Psychol. Behav.*, vol. 61, pp. 107–119, 2019.
- [118] A. Ulahannan *et al.*, "User expectations of partial driving automation capabilities and their effect on information design preferences in the vehicle," *Appl. Ergonom.*, vol. 82, p. 102969, 2020.
- [119] C. Strauch *et al.*, "Real autonomous driving from a passengers perspective: Two experimental investigations using gaze behaviour and trust ratings in field and simulator," *Transp. Res. Part F: Traffic Psychol. Behav.*, vol. 66, pp. 15–28, 2019.
- [120] C. Di Loreto *et al.*, "Science Arts & Métiers (SAM) Real car versus driving simulator comparison of head dynamics in emergency braking events," in *Proc. Driving Simul. Conf. Exhib.*, Strasbourg, France, Sep., pp. 51–55. [Online]. Available: <https://hal.archives-ouvertes.fr/hal-02293336>
- [121] J. Rodriguez, P. Freeman, J. Wagner, W. Bridges, P. Pidgeon, and K. Alexander, "Automotive steering system preferences evaluated using a driving simulator," *Int. J. Automot. Technol.*, vol. 17, no. 1, pp. 71–81, 2016.
- [122] Y. Shida, H. Okajima, D. Matsuno, and N. Matsunaga, "Evaluation of steering model depending on gazing distance by using driving simulator," in *Proc. 16th Int. Conf. Control, Autom. Syst.*, 2016, pp. 39–44.
- [123] Y. E. Papelis, G. S. Watson, and T. L. Brown, "An empirical study of the effectiveness of electronic stability control system in reducing loss of vehicle control," *Accident Anal. Prevention*, vol. 42, no. 3, pp. 929–934, 2010.
- [124] J. Yoon, W. Cho, J. Kang, B. Koo, and K. Yi, "Design and evaluation of a unified Chassis control system for rollover prevention and vehicle stability improvement on a virtual test track," *Control Eng. Pract.*, vol. 18, no. 6, pp. 585–597, 2010.
- [125] A. Emadi, "Transportation 2.0," *IEEE Power and Energy Mag.*, vol. 9, no. 4, pp. 18–29, Jul./Aug. 2011.
- [126] L. Serrao, S. Onori, and G. Rizzoni, "A comparative analysis of energy management strategies for hybrid electric vehicles," *J. Dyn. Syst., Meas., Control*, vol. 133, no. 3, 2011.
- [127] A. Emadi, *Advanced Electric Drive Vehicle*. Boca Raton, FL: CRC Press, Oct. 2014.
- [128] Q. Jiang, F. Ossart, and C. Marchand, "Comparative study of real-time hev energy management strategies," *IEEE Trans. Veh. Technol.*, vol. 66, no. 12, pp. 10 875–10 888, Dec. 2017.
- [129] J. Wang and D. Söffker, "Improving driving efficiency for hybrid electric vehicle with suitable interface," in *Proc. IEEE Int. Conf. Syst., Man, Cybern.*, 2016, pp. 000 928–000 933.
- [130] C. P. Rommerskirchen, M. Helmbrecht, and K. J. Bengler, "The impact of an anticipatory eco-driver assistant system in different complex driving situations on the driver behavior," *IEEE Intell. Transp. Syst. Mag.*, vol. 6, no. 2, pp. 45–56, Summer 2014.
- [131] S. Jones *et al.*, "V2x based traffic light assistant for increased efficiency of hybrid & electric vehicles," in *Proc. AmE 2016-Automot. Meets Electron.; 7th GMM-Symp. VDE*, 2016, pp. 1–5.
- [132] S. M. Pampel, S. L. Jamson, D. L. Hibberd, and Y. Barnard, "How i reduce fuel consumption: An experimental study on mental models of eco-driving," *Transp. Res. Part C: Emerg. Technol.*, vol. 58, pp. 669–680, 2015.
- [133] Y. Wu, X. Zhao, J. Rong, and Y. Zhang, "How eco-driving training course influences driver behavior and comprehensibility: A driving simulator study," *Cognit., Technol. Work*, vol. 19, no. 4, pp. 731–742, 2017.

- [134] A. Jamson, D. L. Hibberd, and N. Merat, "The design of haptic gas pedal feedback to support eco-driving," in *Proc. 7th Int. Driving Symp. Hum. Factors Driver Assessment, Training, Veh. Des.*, Univ. Iowa, 2013, pp. 264–270.
- [135] D. Kok, M. Knowles, and A. Morris, "Building a driving simulator as an electric vehicle hardware development tool," in *Proc. Driving Simul. Conf.*, Paris, France, Sep. 2012, pp. 1–7.
- [136] D. H. Phuc, P. Rasincharensak, and M. Nagai, "Study on driver model for hybrid truck based on driving simulator experimental results," *IATSS Res.*, vol. 42, no. 1, pp. 18–23, 2018.
- [137] E. Baumgartner, A. Ronellenfitch, H.-C. Reuss, and D. Schramm, "Using a dynamic driving simulator for perception-based powertrain development," *Transp. Res. Part F: Traffic Psychol. Behav.*, vol. 61, pp. 281–290, 2019.
- [138] X. Wang, X. Wang, B. Cai, and J. Liu, "Combined alignment effects on deceleration and acceleration: a driving simulator study," *Transp. Res. Part C: Emerg. Technol.*, vol. 104, pp. 172–183, 2019.
- [139] A. J. Filtner et al., "Safety implications of co-locating road signs: A driving simulator investigation," *Transp. Res. Part F: Traffic Psychol. Behav.*, vol. 47, pp. 187–198, 2017.
- [140] M.-W. Kang and S. U. Momtaz, "Assessment of driver compliance on roadside safety signs with auditory warning sounds generated from pavement surface—a driving simulator study," *J. Traffic Transp. Eng. (English edition)*, vol. 5, no. 1, pp. 1–13, 2018.



LUCAS BRUCK (Student Member, IEEE) received the B.S. degree in mechanical engineering from the Federal University of Sao Joao del-Rei, Brazil (UFSJ), in 2014. In 2018, he was with the McMaster Automotive Resource Centre (MARC), McMaster University as a M.S. Student. Lucas is currently working toward the Ph.D. degree with the same university. In the following five years, he gathered experience in industry working with vehicle dynamics, virtual simulation, and testing. His main field of expertise is vehicle dynamics,

active controls and safety systems.



BRUCE HAYCOCK is a Scientist and the Lead Engineer for real-time computing with the Challenging Environment Assessment Laboratory. He is an Assistant Professor with the University of Toronto Institute for Aerospace Studies. Bruce's background and research interests focus is primarily human-in-the-loop vehicle simulation, including simulator technologies, vehicle dynamics, and human control of vehicles. This experience is now being applied at the unique simulation facilities, TRI to enable a broad range of research, notably

through the development and optimization of DriverLab and control of the CEAL hardware.



ALI EMADI (Fellow, IEEE) received the B.S. and M.S. degrees in electrical engineering with highest distinction from the Sharif University of Technology, Tehran, Iran, in 1995 and 1997, respectively, and the Ph.D. degree in electrical engineering from Texas A&M University, College Station, TX, USA, in 2000. He is the Canada Excellence Research Chair Laureate with McMaster University, Hamilton, Ontario, Canada. He is also the holder of the NSERC/FCA Industrial Research Chair in electrified powertrains and Tier I Canada Research Chair

in transportation electrification and smart mobility. Before joining McMaster University, Dr. Emadi was the Harris Perlstein Endowed Chair Professor of Engineering and Director of the Electric Power and Power Electronics Center and Grainger Laboratories with the Illinois Institute of Technology in Chicago, where he established research and teaching facilities as well as courses in power electronics, motor drives, and vehicular power systems. He was the Founder, Chairman, and President of Hybrid Electric Vehicle Technologies, Inc. (HEVT) - a university spin-off company of Illinois Tech. Currently, he is the President and Chief Executive Officer with Enedym Inc. and Menlob Inc. two McMaster University spin-off companies. He is the Principal Author/Co-Author of over 500 journal and conference papers as well as several books including *Vehicular Electric Power Systems* (2003), *Energy Efficient Electric Motors* (2004), *Uninterruptible Power Supplies and Active Filters* (2004), *Modern Electric, Hybrid Electric, and Fuel Cell Vehicles* (2nd ed, 2009), and *Integrated Power Electronic Converters and Digital Control* (2009). He is also the Editor of the Handbook of *Automotive Power Electronics and Motor Drives* (2005) and *Advanced Electric Drive Vehicles* (2014). He is the Co-Editor of the *Switched Reluctance Motor Drives* (2018). Dr. Emadi was the Inaugural General Chair of the 2012 IEEE Transportation Electrification Conference and Expo (ITEC) and has chaired several IEEE and SAE conferences in the areas of vehicle power and propulsion. He was the founding Editor-in-Chief of the IEEE TRANSACTIONS ON TRANSPORTATION ELECTRIFICATION from 2014 to 2020.